

**CMPE 491**  
**Senior Project I**  
**Project Specifications Report**



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## 1. Introduction

### 1.1. Description

This project aims to develop an AI-based camera image enhancement system capable of improving the visual quality of low-light, noisy, or low-resolution images. Unlike traditional filtering techniques that rely on fixed rules, this system uses deep learning models that learn directly from data. Through this adaptive structure, the model can brighten dark environments, reduce visual noise, sharpen blurred regions, and correct color imbalances automatically.

The system is designed to be both efficient and sustainable. By incorporating optimized model architectures and energy-aware processing strategies, the project minimizes resource consumption while maintaining high-quality output. Built on an open-source foundation, the solution supports collaboration, long-term adaptability, and transparent development. With its flexible and data-driven nature, the system targets various real-world applications, including smart city surveillance, autonomous driving, healthcare imaging, environmental monitoring, and industrial quality control.

### 1.2. Constraints

During development, several constraints must be taken into account:

#### Technical Constraints

- **Computational Cost:** Training deep learning models requires high GPU power and long training times, which increases development cost and limits experimentation speed.
- **Real-Time Performance Challenges:** High-resolution images require optimized processing. Without proper model compression or architectural optimization, latency may occur.
- **Hardware Compatibility:** Different deployment environments (embedded systems, servers, local machines) vary in performance, making universal optimization difficult.
- **Dataset Limitations and Labeling Constraints:** Since the system is designed for image enhancement rather than classification, the project does not rely on manually labeled datasets. Instead, it requires large paired or unpaired datasets consisting of high-quality and low-quality image samples.

This introduces several important constraints:

- High-quality paired datasets are difficult to collect and may require controlled environments.

- Unpaired datasets (used in unsupervised enhancement techniques) may introduce inconsistencies in model learning.
- The absence of traditional labels reduces annotation cost, but makes it harder to quantitatively measure ground-truth accuracy.
- Ensuring dataset variety (lighting, noise types, camera sensors, environments) is challenging but necessary for preventing model bias and improving generalization.
- Ethical and legal considerations arise when datasets contain real-world imagery, requiring proper anonymization or publicly licensed datasets.

### **Environmental and Sustainability Constraints**

- Training and fine-tuning models consume significant energy.
- Efficient architectures are needed to reduce environmental impact and operational cost.

### **Legal and Ethical Constraints**

- Image datasets must comply with data protection rules (e.g., GDPR).
- Any dataset containing people or identifiable scenes must undergo anonymization or use licensed sources.
- Fairness concerns arise if the dataset overrepresents certain environments, lighting conditions, or camera types.

### **Operational Constraints**

- The system must integrate with existing camera pipelines without requiring major hardware changes.
- Consistency must be maintained across different image types and environmental conditions.
- Long-term maintainability depends on a modular, open-source structure and clear documentation.

### 1.3. Professional and Ethical Issues

Developing a system that processes visual data introduces several professional and ethical responsibilities:

- **Data Privacy & Security:**

All image data must be handled with strict confidentiality. Secure storage, encrypted transmission, and controlled access are essential.

- **Fairness & Bias Prevention:**

The system must avoid producing biased or misleading outputs. Training data and model behavior should be monitored to ensure fairness across diverse conditions.

- **Transparency & Explainability:**

Users should have clear information regarding how the model processes images and what types of data are used. Transparency builds trust and supports ethical deployment.

- **Regulatory Compliance:**

Adherence to international and national data protection laws, AI ethics guidelines, and domain-specific regulations is required throughout development.

- **Responsible Deployment:**

In sensitive domains like healthcare or surveillance, the system should undergo thorough testing to minimize the risk of misinterpretation or harm.

## 2. Requirements

### Functional Requirements

- The system shall enhance images captured under low-light, noisy, or low-resolution conditions.
- It shall automatically adjust brightness, sharpness, noise levels, and color balance.
- It shall support common image file formats (e.g., JPEG, PNG, TIFF).
- Users shall be able to process images individually or in batches.
- The system shall provide visibly improved image quality without requiring manual tuning.

### Non-Functional Requirements

- **Performance:** The system must operate efficiently without excessive resource consumption.
- **Compatibility:** It must run on a variety of hardware configurations and operating environments.
- **Modularity:** The architecture should support easy updates, improvements, and component replacement.
- **Security:** User data must be processed with strong privacy protection and secure handling.
- **Maintainability:** Documentation and code structure should support long-term development efforts.
- **Scalability:** The system should allow the integration of improved or alternative AI models in the future.

### 3. References

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
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- ISO/IEC 23091-4:2020 – Image Coding Systems and Quality Evaluation Metrics.